

Third Assignment: Primal–Dual Bounds and Branch-and-Bound Algorithm

Assignment Solutions

Instructions (from original assignment)

- Answer conceptual questions clearly and precisely.
- Show all reasoning in computational parts.
- For algorithmic parts, provide:
 - A flowchart
 - Clear pseudocode

1 Primal and Dual Bounds

Conceptual Questions

1. What is a primal bound in a minimization problem?

In a minimization problem, a primal bound is a valid upper limit on the optimal objective value. It is obtained by evaluating the objective function at any feasible solution to the problem. The true minimum will be less than or equal to this bound.

Source: Rao, S. S. (2020). Engineering Optimization: Theory and Practice (5th ed.).

2. What is a dual bound?

A dual bound is a valid lower limit (for a minimization problem) or upper limit (for a maximization problem) on the optimal objective value. It is typically found by relaxing the problem (e.g., ignoring integrality constraints or using Lagrangian relaxation), representing the theoretical limit of how good the objective could possibly be.

Source: Nocedal, J., & Wright, S. J. (2006). Numerical Optimization (2nd ed.).

3. Why is a dual bound always optimistic (for minimization)?

For a minimization problem, a dual bound is optimistic because the relaxation expands the feasible region or removes penalties. This allows the solver to find a solution that is potentially “cheaper” than what is actually achievable under the strict original constraints.

Source: Nocedal, J., & Wright, S. J. (2006). Numerical Optimization (2nd ed.).

4. Define the optimality gap.

The optimality gap is the absolute or relative difference between the best known primal bound

(the best feasible solution found so far) and the best known dual bound (the theoretical limit).
Source: Iqbal, K. (2013). Fundamental Engineering Optimization Methods.

5. **When can we certify optimality?**

Optimality is certified when the primal bound exactly equals the dual bound (i.e., the optimality gap is zero). This proves that no better solution can possibly exist within the feasible space.

Source: Bonnans, J. F., et al. (2006). Numerical Optimization: Theoretical and Practical Aspects.

2 LP Relaxation

Consider the following integer program:

$$\max 5x_1 + 4x_2$$

subject to

$$6x_1 + 4x_2 \leq 24, \quad x_1, x_2 \in \mathbb{Z}_{\geq 0}.$$

1. **What is an LP relaxation of an IP/MILP?**

An LP relaxation is the linear programming problem obtained by dropping the integrality constraints from an Integer Program (IP) or Mixed-Integer Linear Program (MILP), allowing the variables to take continuous fractional values.

Source: Rao, S. S. (2020). Engineering Optimization: Theory and Practice.

2. **Solve the LP relaxation geometrically:**

Relaxing the constraints gives continuous bounds $x_1, x_2 \geq 0$. The feasible region is bounded by the axes and the line $6x_1 + 4x_2 = 24$. The vertices of this triangular region are:

- $(0, 0) \Rightarrow Z = 0$
- $(4, 0) \Rightarrow Z = 5(4) + 4(0) = 20$
- $(0, 6) \Rightarrow Z = 5(0) + 4(6) = 24$

The optimal continuous solution is $x_1 = 0, x_2 = 6$ with $Z_{LP} = 24$.

Source: Applied geometric method from Rao, S. S. (2020).

3. **Use it to compute a dual bound.**

Because this is a maximization problem, the LP relaxation provides a valid upper bound. The dual bound is 24.

Source: Iqbal, K. (2013). Fundamental Engineering Optimization Methods.

4. **Find a feasible integer solution and compute a primal bound.**

The optimal LP-relaxation solution ($x_1 = 0, x_2 = 6$) already consists entirely of integers. Therefore, it is feasible for the original IP. The primal bound is 24.

Source: Rao, S. S. (2020). Engineering Optimization: Theory and Practice.

5. **Compute the optimality gap.**

$$\text{Optimality Gap} = |\text{Dual Bound} - \text{Primal Bound}| = |24 - 24| = 0.$$

Source: Bonnans, J. F., et al. (2006). Numerical Optimization.

3 Duality in Linear Programming

Conceptual Questions

1. **What is the dual problem of a linear program?**

For every linear program (the primal), there is an associated linear program called the dual. Every variable in the primal becomes a constraint in the dual, and every constraint in the primal becomes a variable in the dual. The dual bounds the optimal value of the primal.

Source: Nocedal, J., & Wright, S. J. (2006). Numerical Optimization.

2. **State the Weak Duality Theorem.**

For any feasible solution x to the primal problem and any feasible solution y to the dual problem, the objective value of the primal is bounded by the objective value of the dual (e.g., $c^T x \leq b^T y$ for primal maximization).

Source: Rao, S. S. (2020). Engineering Optimization: Theory and Practice.

3. **What does Weak Duality imply about bounds?**

It implies that any feasible solution to the dual provides a valid dual bound (upper bound for maximization, lower bound for minimization) for the primal problem.

Source: Iqbal, K. (2013). Fundamental Engineering Optimization Methods.

4. **State the Strong Duality Theorem.**

If either the primal or dual problem has a finite optimal solution, then both have finite optimal solutions, and their optimal objective values are equal ($Z^* = W^*$).

Source: Nocedal, J., & Wright, S. J. (2006). Numerical Optimization.

5. **What is complementary slackness (conceptually)?**

At optimality, if a constraint in one problem is strictly satisfied (has non-zero slack), the corresponding variable in the paired problem must be zero. Conversely, if a variable is strictly positive, its corresponding constraint must be binding (zero slack).

Source: Rao, S. S. (2020). Engineering Optimization.

Exercise

Consider the following LP:

$$\max 3x_1 + 2x_2$$

subject to

$$x_1 + x_2 \leq 4, \quad 2x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0.$$

1. **Write the dual problem.**

Minimize $W = 4y_1 + 5y_2$

subject to

$$y_1 + 2y_2 \geq 3, \quad y_1 + y_2 \geq 2, \quad y_1, y_2 \geq 0.$$

Source: Formulation based on Nocedal & Wright (2006).

2. **Solve the primal geometrically.**

Intersection of $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5$ is at $(1, 3)$. Vertices are $(0, 0)$, $(2.5, 0)$, $(1, 3)$,

(0, 4). Evaluating $Z = 3x_1 + 2x_2$:

$$Z(0, 4) = 8, \quad Z(1, 3) = 9, \quad Z(2.5, 0) = 7.5.$$

Thus the maximum is $Z^* = 9$ at $x_1 = 1, x_2 = 3$.

Source: Rao, S. S. (2020). Engineering Optimization.

3. Solve the dual.

Intersection of $y_1 + 2y_2 = 3$ and $y_1 + y_2 = 2$ is at (1, 1). Vertices are (0, 2), (1, 1), (3, 0). Evaluating $W = 4y_1 + 5y_2$:

$$W(0, 2) = 10, \quad W(1, 1) = 9, \quad W(3, 0) = 12.$$

Thus the minimum is $W^* = 9$ at $y_1 = 1, y_2 = 1$.

Source: Rao, S. S. (2020). Engineering Optimization.

4. Verify strong duality.

The optimal primal objective value $Z^* = 9$ equals the optimal dual objective value $W^* = 9$, verifying Strong Duality.

Source: Nocedal, J., & Wright, S. J. (2006).

4 Bounding Techniques

Conceptual Questions

1. Why does LP relaxation provide a valid bound?

By dropping integrality constraints, the feasible region is enlarged. Because the solver optimizes over a larger set of possibilities, the continuous optimum will always be greater than or equal to (for maximization) the restricted integer optimum.

Source: Iqbal, K. (2013). Fundamental Engineering Optimization Methods.

2. What is a Lagrangian relaxation (conceptually)?

A technique where hard constraints are removed from the constraint space and moved into the objective function, penalized by Lagrangian multipliers if they are violated.

Source: Nocedal, J., & Wright, S. J. (2006).

3. What other general methods provide a dual bound?

Surrogate Relaxation: Aggregates multiple constraints into a single constraint using weighted sums.

Combinatorial Relaxations: Uses graph/network properties (e.g., Minimum Spanning Tree bounds for TSP variants).

Source: Rao, S. S. (2020). Engineering Optimization.

4. What makes a bound “strong”?

A bound is strong if it is very close to the true integer optimal value, creating a tight optimality gap. Strong bounds allow branch-and-bound to prune unfruitful search trees faster.

Source: Bonnans, J. F., et al. (2006).

5. Comparison Table

(Kept as text, like the PDF)

Approach	Use Case	Bound Quality	Properties
LP Relaxation	IP, MILP	Variable (often weak)	Fast to compute, drops integrality
Lagrangian	Constrained IP	Strong	Good for specific block structures
Surrogate IP	Moderate to Strong	Reduces constraints to a single row	

Source: *General taxonomy derived from Rao, S. S. (2020).*

Exercise (Knapsack)

For the knapsack problem:

$$\max \sum_{i=1}^n p_i x_i \quad \text{subject to} \quad \sum_{i=1}^n w_i x_i \leq W, \quad x_i \in \{0, 1\}.$$

Given profits $p = (10, 8, 6)$, weights $w = (5, 4, 3)$, capacity $W = 7$.

1. Greedy algorithm pseudocode (LP relaxation / fractional knapsack).

The algorithm sorts items by value-to-weight ratio and picks them until full.

1. For each item i , compute $\text{ratio}[i] = p[i]/w[i]$.
2. Sort items in descending order of ratio.
3. $\text{TotalValue} = 0$, $\text{RemainingW} = W$.
4. For each item i in sorted list:
5. If $w[i] \leq \text{RemainingW}$:
6. $\text{TotalValue} += p[i]$, $\text{RemainingW} -= w[i]$.
7. Else:
8. $\text{TotalValue} += p[i] \cdot (\text{RemainingW}/w[i])$.
9. **Break**
10. Return TotalValue .

Source: *Iqbal, K. (2013).*

2. How does the fractional knapsack provide an upper bound?

Because it optimizes an LP relaxation: allowing fractional items can only improve (or match) the best 0/1 solution, thus giving an upper bound for the integer knapsack.

Source: *Rao, S. S. (2020).*

3. What is the integrality gap?

The integrality gap is the difference (or ratio) between the optimal objective value of the Integer Program and the optimal objective value of its continuous LP relaxation.

Source: *Bonnans, J. F., et al. (2006).*

4. Compute the dual bound (greedy).

Ratios: $10/5 = 2$, $8/4 = 2$, $6/3 = 2$ (all identical). Take Item 1 entirely: weight 5, profit 10. Remaining capacity: $7 - 5 = 2$. Take a fraction of Item 2: fraction $2/4 = 0.5$, profit $0.5 \times 8 = 4$. Total dual bound = $10 + 4 = 14$.

Source: *Calculation applied via Iqbal, K. (2013).*

5 Branch and Bound: An Exact Algorithm

Conceptual Questions

1. **What is an exact algorithm? When is it used?**

An exact algorithm guarantees finding the absolute mathematical optimum and proving its optimality in finite time. It is used when approximation or heuristic solutions are unacceptable due to safety, financial, or strict constraint requirements.

Source: Rao, S. S. (2020).

2. **What is the main idea of Branch and Bound?**

It uses “divide and conquer”: partition the feasible region into subproblems (branching) and use relaxations to compute bounds (bounding), discarding nodes that cannot improve the current best integer solution.

Source: Nocedal, J., & Wright, S. J. (2006).

3. **What are the three main components?**

(1) Branching, (2) Bounding, (3) Fathoming (Pruning).

Source: Rao, S. S. (2020).

4. **What is fathoming?**

Fathoming is permanently closing a node if (i) its LP relaxation is infeasible, (ii) its bound is worse than the incumbent, or (iii) the LP solution is already integer.

Source: Iqbal, K. (2013).

Exercise

Solve:

$$\max 3x_1 + 2x_2$$

subject to

$$2x_1 + x_2 \leq 4, \quad x_1, x_2 \in \mathbb{Z}_{\geq 0}.$$

1. **Tree, bounds, pruning, and optimal solution (as in the PDF).**

Node 0 (Root): Solve continuous LP relaxation. The constraint is $2x_1 + x_2 \leq 4$. Vertices: $(0, 0)$, $(2, 0)$, $(0, 4)$.

At $(2, 0)$: $Z = 3(2) + 2(0) = 6$.

At $(0, 4)$: $Z = 3(0) + 2(4) = 8$.

So $Z_{LP} = 8$ at $(x_1, x_2) = (0, 4)$. This solution is already integer, so it is optimal for the IP. Node 0 is pruned by integrality.

Optimal Solution: $x_1 = 0$, $x_2 = 4$ with $Z^* = 8$. The search tree consists solely of Node 0.

Source: Calculated using principles from Rao, S. S. (2020).

Flowchart sequence (text version)

Flowchart sequence for maximization:

1. Initialization: Set incumbent $Z^* = -\infty$, add root node to active list.

2. Termination: If active list empty $\rightarrow$ return Z^* ; else continue.
 3. Node selection: Pop a node N from active list.
 4. LP relaxation: Solve continuous LP for node N .
 5. Bounding test (feasibility): If LP infeasible $\rightarrow$ prune (fathom), go to 2.
 6. Bounding test (optimality): If $Z_{LP} \leq Z^* \rightarrow$ prune by bound, go to 2.
 7. Integrality test: If LP solution is integer:
 - Update $Z^* = Z_{LP}$, prune by integrality, go to 2.
- Otherwise branch:
- Choose fractional variable $x_i = v$.
 - Create child 1: $x_i \leq \lfloor v \rfloor$ and child 2: $x_i \geq \lceil v \rceil$.
 - Add children to active list, go to 2.

Pseudocode

```

Function BranchAndBound_Max():
    Best_Z = -infinity
    Best_X = null
    Active_Nodes = {Root_Problem}

    While Active_Nodes is not empty:
        Current_Node = Select_And_Remove(Active_Nodes)
        LP_Solution, LP_Z = Solve_LP_Relaxation(Current_Node)

        If LP_Solution is INFEASIBLE:
            Continue // Fathom by infeasibility

        If LP_Z <= Best_Z:
            Continue // Fathom by bound

        If Is_Integer(LP_Solution):
            Best_Z = LP_Z
            Best_X = LP_Solution
            Continue // Fathom by integrality

        // Branching
        Var_x = Choose_Fractional_Variable(LP_Solution)
        Child_1 = Create_Node(Current_Node, Var_x <= Floor(Var_x.value))
        Child_2 = Create_Node(Current_Node, Var_x >= Ceil(Var_x.value))
        Active_Nodes.add(Child_1)
        Active_Nodes.add(Child_2)

    Return Best_X, Best_Z

```

Source: Standard Branch and Bound architecture as defined in Nocedal & Wright (2006) and Rao (2020).